

KARTIKEYA GAUR

Centre for Infrastructure, Sustainable Transportation and Urban Planning (CiSTUP)
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EDUCATION

Indian Institute of Science (IISc) <i>M.Tech. in Smart Mobility and Logistics Systems</i>	Bengaluru, India 2027
National Institute of Technology (NIT) Hamirpur <i>B.Tech. in Civil Engineering</i>	Himachal Pradesh, India 2024

RESEARCH & PROFESSIONAL EXPERIENCE

University of New South Wales (UNSW) <i>Research Intern</i>	Sydney, Australia Jun – Jul 2026
Developed a comparative route-choice modeling framework for public transit networks in Sydney and Bengaluru, implementing Expanded Path Size Logit (EPS) and Nested Recursive Logit (NRL) using real GTFS and ridership data under Prof. Taha Rashidi. [Report] [GitHub]	
Indian Institute of Remote Sensing (IIRS), ISRO <i>Junior Research Fellow</i>	Dehradun, India Sept 2024 – July 2025
- Developed an automated framework for river discharge and bathymetry estimation by integrating multi-mission satellite altimetry (ICESat-2, SWOT, Sentinel-3) using DBSCAN clustering and Piecewise NLSB algorithms. - Conducted extensive ADCP-based field validation across 12 Ganga River sites, validating rating curves and achieving a bathymetry prediction accuracy (RMSE) of 2.40 m. [Report]	
BISAG-N <i>Geospatial Project Trainee</i>	Gandhinagar, India Dec 2022 – Jan 2023
- Proposed three railway alignments in the mountainous terrain of Northeast India under the PM GatiShakti National Master Plan. - Analyzed terrain characteristics and topology in QGIS to generate shapefiles and model discharge points to support check-dam planning. [Report]	
Defence Laboratory (DRDO) <i>Technical Trainee</i>	Jodhpur, India Jun – Jul 2022
- Evaluated military trafficability in arid regions of Jaisalmer under the supervision of Dr. S.L. Borana. - Combined supervised LULC classifications, DEM derivatives, and digitized soil maps in ArcGIS to generate a weighted overlay trafficability map. [Report]	

PROJECTS

Scalable Algorithm for Public Transit Skims <i>Supervisors: Prof. Tarun Rambha & Prof. Abdul R. Pinjari</i>	<i>M.Tech Thesis</i>
Contributing to a scalable C++ routing framework designed to generate transit network skim matrices, aiming to simulate millions of agent journeys efficiently for Bengaluru.	
Implementation of a Convex Travel Demand Forecasting Model	<i>Choice Modelling</i>
- Reproduced a combined travel demand optimization model (trip distribution, modal split, and traffic assignment) as a single convex program using Julia/JuMP. - Reformulated conditional entropy terms into exponential cones and link-latency integrals into power cones for efficient resolution using the MOSEK solver. [Report]	
Semiring-Based Dijkstra for Multimodal Route Planning	<i>Algorithms & Network Science</i>
Designed an algebraic routing engine in Python that unifies road and BMTC bus networks into a single graph and computes optimal paths with a generalized Dijkstra algorithm over interchangeable semirings (distance, time, fare, safety, and lexicographic combinations). [Report]	
Nighttime Street Light Audit Using Computer Vision	<i>Edge AI</i>
Built a nighttime streetlight fault detector (ON/OFF) as a team of four: collected dashboard-camera footage over 16.6 km of Bengaluru roads, annotated ~2.1k frames, and used CLAHE, gamma correction, and unsharp masking to handle low light and motion blur. YOLOv5nu reached 0.886 mAP50 with 2.5M parameters and 3.2 ms GPU inference. [Report] [GitHub]	

- Compressed the models with FP16/INT8 quantization, one-shot and iterative pruning (up to 50% sparsity), and knowledge distillation, then exported at 320×320 (ONNX, TFLite INT8) for a smartphone web app with on-device inference.

Last-Mile Connectivity Analysis and Visualization

Multi-Modal Mobility Systems

- Analyzed commuter survey data across Bengaluru, Delhi, and Nagpur and built interactive Power BI dashboards; mapped Delhi's **high non-walk dependency zones** (13.35% area, 38/290 wards) and found Nagpur has the highest population accessibility per station (0.88%). [Report]

Intra-Urban Heat Island Effect Analysis: Case Study of Chandigarh

BTech Project

- Leveraged USGS Landsat-8 thermal imagery (2003–2023) in Google Earth Engine to analyze Land Surface Temperature (LST) variations and quantify the Intra-Urban Heat Island effect, observing a long-term temperature rise of 2.80°C in Chandigarh (**0.14°C/year**) and a higher surge of 6.77°C in Jalandhar (**0.34°C/year**). [Report]

Capstone Project: Warehouse Optimization at Land Port Petrapole

Partner: Ernst & Young (EY) LLP

- Formulated physical and process interventions to address a 191.7% warehouse over-utilization rate.
- Modeled a Selective Pallet Racking system using ABC-XYZ inventory classification and proposed a 5-layer IoT data architecture.
- Constructed a discrete-event simulation (DES) using AnyLogic to analyze vessel/vehicle queuing bottlenecks. [Report]

Contributor Verification using Robust Deep Neural Networks

ML for Cyber-Physical Systems

- Built a multi-branch Deep Neural Network (Keras Functional API) with L2 regularization and Dropout for binary classification on 88,340 samples, ranking 11/25 on the private leaderboard (score 0.8607). [Report]

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python (NumPy, Pandas, Scipy), C++, R, Julia/JuMP.
- **Machine Learning & Optimization:** TensorFlow, PyTorch, Keras, Scikit-learn, MOSEK.
- **Geospatial & Design Tools:** ArcGIS, QGIS, GEE (Google Earth Engine), ENVI, AutoCAD.
- **Simulation & Visualization:** SimPy, AnyLogic, Power BI.
- **Languages:** English (fluent), Hindi (native), French (A1).

PUBLICATIONS

- Gaur, K., et al. (2025). “**SWOT Mission with Wide-Swath Altimetry: Observations and Insights into India’s Inland Waterbodies.**” *Journal of the Indian Society of Remote Sensing*.
- “**Nadir and Wide Swath Altimetry for Inland Water Bodies.**” *Technical Report, Indian Institute of Remote Sensing* (2025). [PDF]

ACHIEVEMENTS & HONORS

- **Secured All India Rank (AIR) 10** in GATE 2023 (Geomatics Engineering) with an 819 score. [Score card]

CO-CURRICULAR & LEADERSHIP ROLES

- **Student Auditor**, Literacy Mission, NIT Hamirpur (2022–24), supporting tutoring for 80+ underprivileged children
- **Coordinator**, C-Helix (Nimbus Technical Festival), organized a QGIS workshop for 100+ undergraduates.

WORKSHOPS & CERTIFICATES

- COEAI-SPARC Workshop on Hybrid Physics-AI Models for Climate, Weather and Water, IIT Kharagpur, June 2024.
- High-End Workshop (Karyashala) on “Mathematical Foundations for AI/ML,” SERB, IIT Indore, MP, May 2024.
- Remote Sensing and GIS in Rural Development, NPTEL, January 2023. [Certificates]